Statistical Modeling in R

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About Me

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- Workshop facilitator
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PART 1

Foundation

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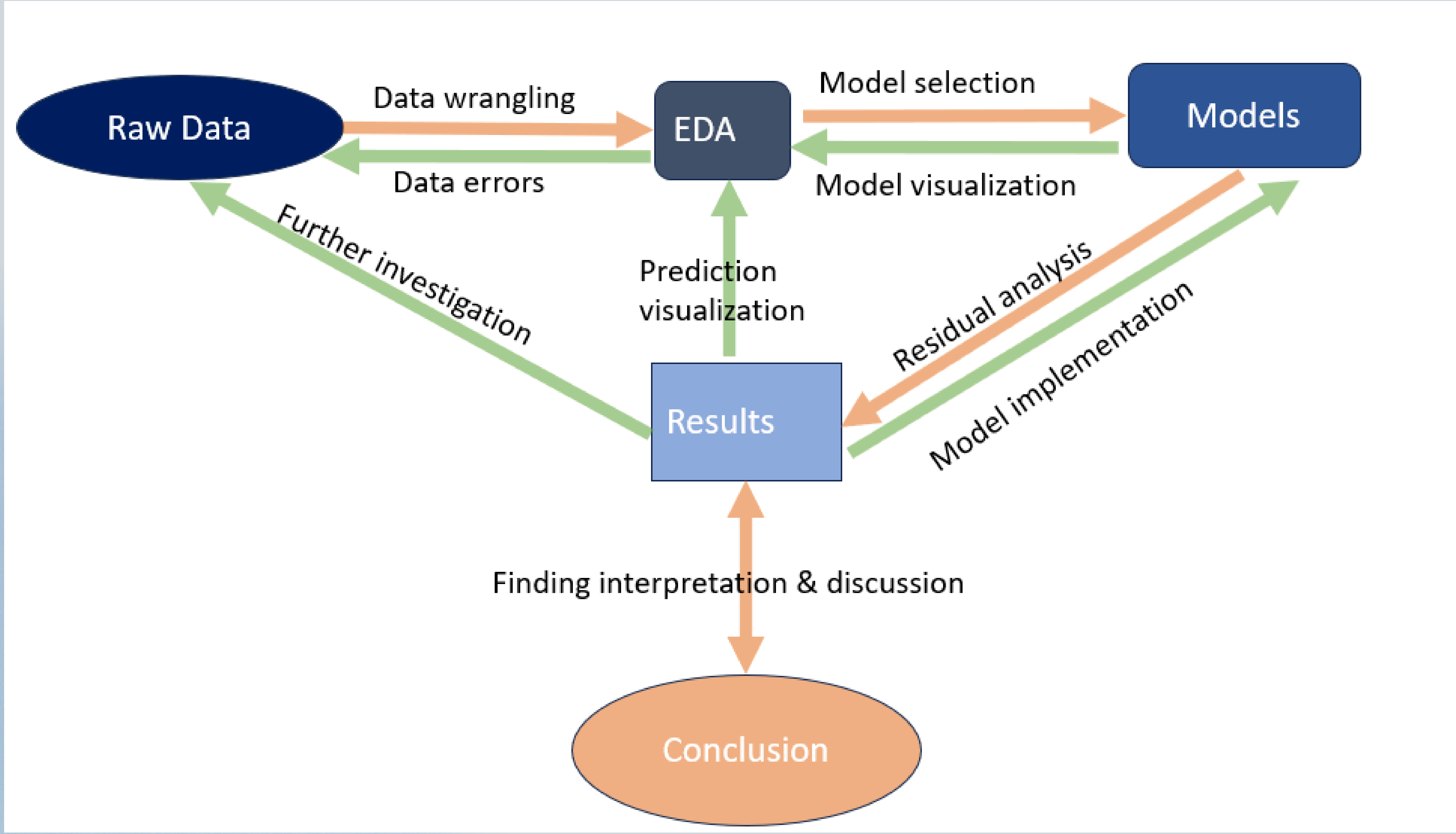
Workshop Overview

01 Data Analysis Workflow

02 Common Statistical Models

Data Analysis Workflow



Why learn statistical modeling?

Advantages of Statistical Models

INTERPRETABILITY

Statistical models often directly show how each variable influences the outcome(e.g., coefficients, significance).

This makes them ideal for explaining results to stakeholders and supporting scientific conclusions.

BETTER FOR DATA QUALITY & DIAGNOSTICS

Statistical models provide built-in tools (residuals, assumptions checks) to detect:

- data issues
- model misspecification

This helps improve both data quality and model reliability.

EFFICIENT WITH SMALLER & STRUCTURED DATA

Statistical models perform well even with:

- smaller datasets
- well-designed experiments

Unlike many ML models, they don't require large amounts of data or heavy computation.

It's always good to know more tools in the data world!

Common Statistical Models

01 Linear Model (LM)

02 Generalized Linear Model (GLM)

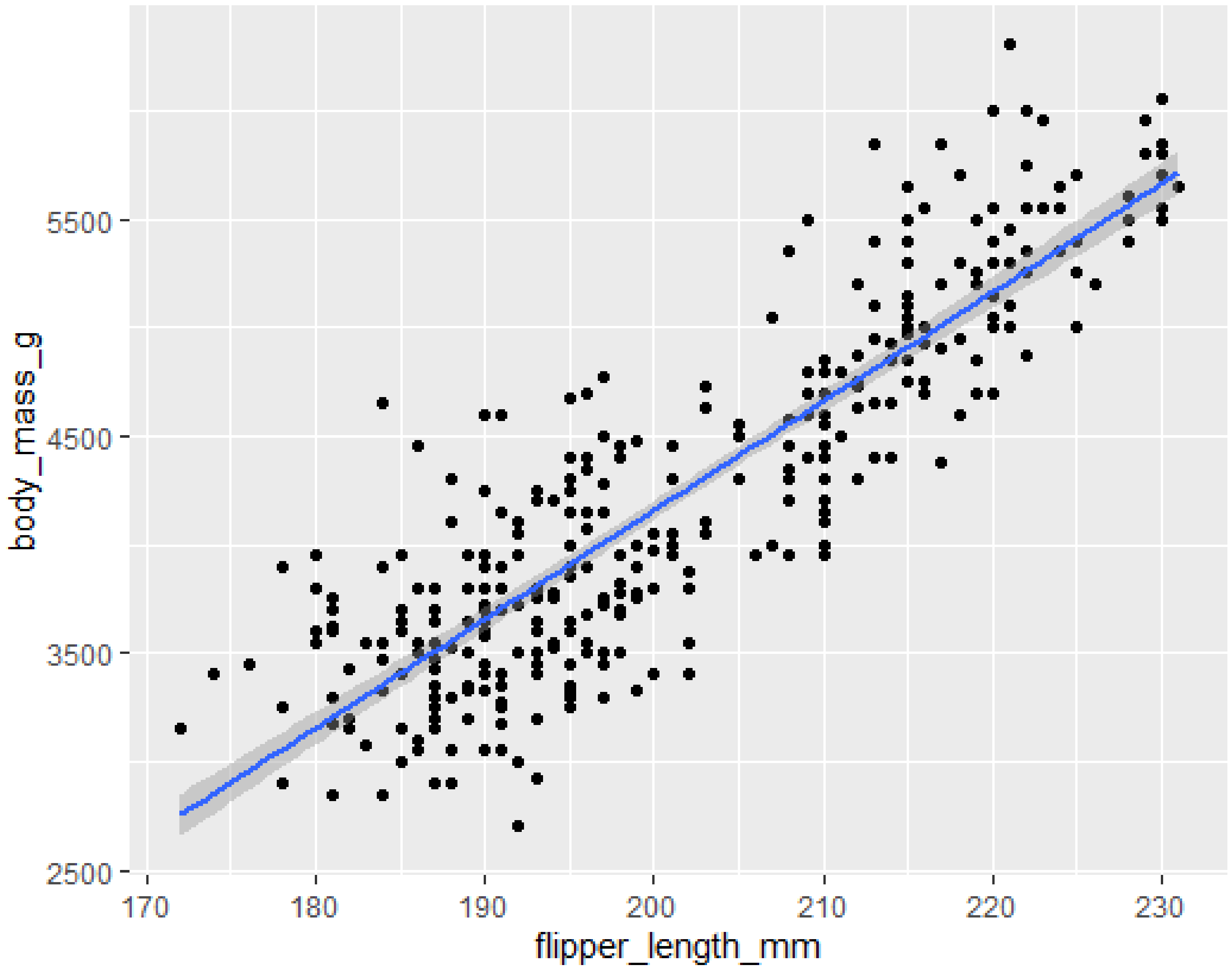
03 Generalized Additive Model (GAM)

04 Generalized Linear Mixed Model (GLMM)

05 Generalized Additive Mixed Model (GAMM)

Linear Model (LM)

$$y = X\beta + \varepsilon, y \sim N(\mu, \sigma^2), \beta = C$$



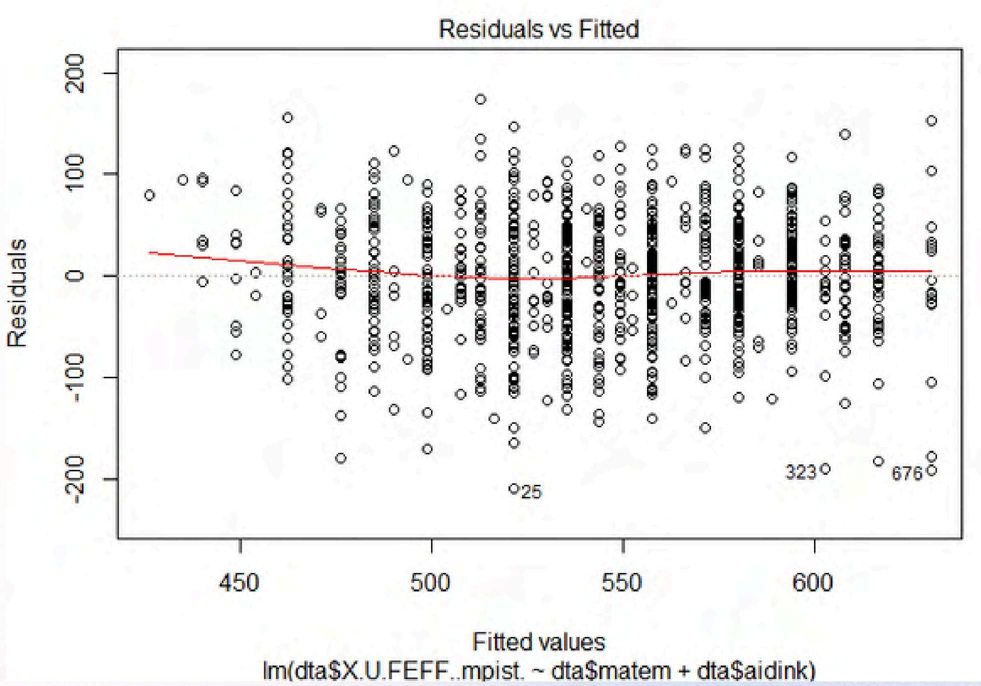
Key Assumptions

- **Linearity: The relationship between input and output is linear.**
- **Homogeneity of variance (Homoscedasticity): The error variance is constant across predictors.**
- **Independence: Observations are independent.**
- **Normality: The response and residuals are normally distributed**

LM Rsidual Analaysis

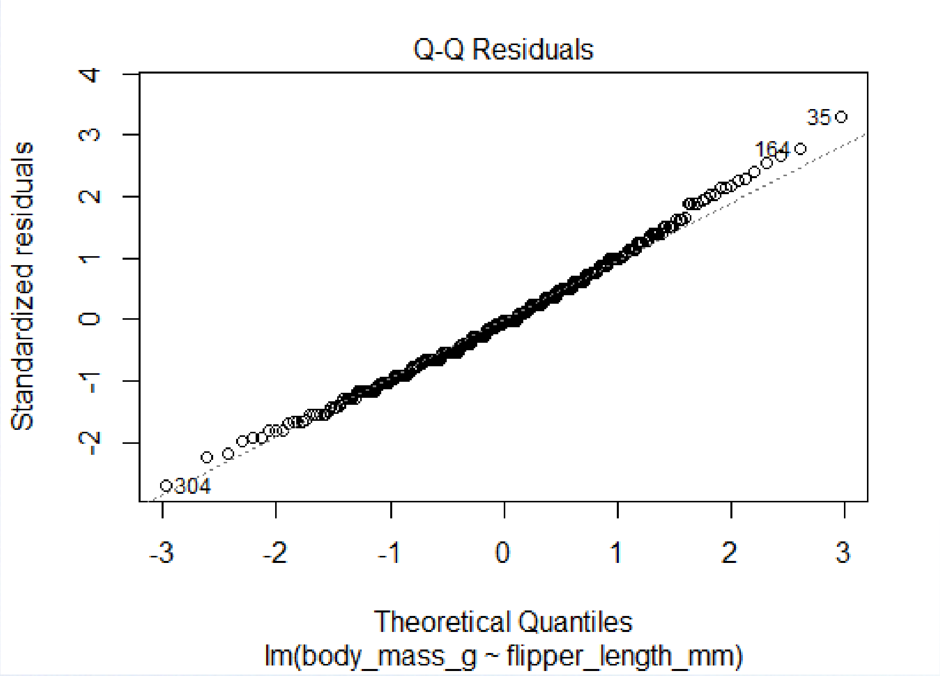
Linearity → Residuals vs Fitted

a flat horizontal line at zero with no distinct pattern is ideal.



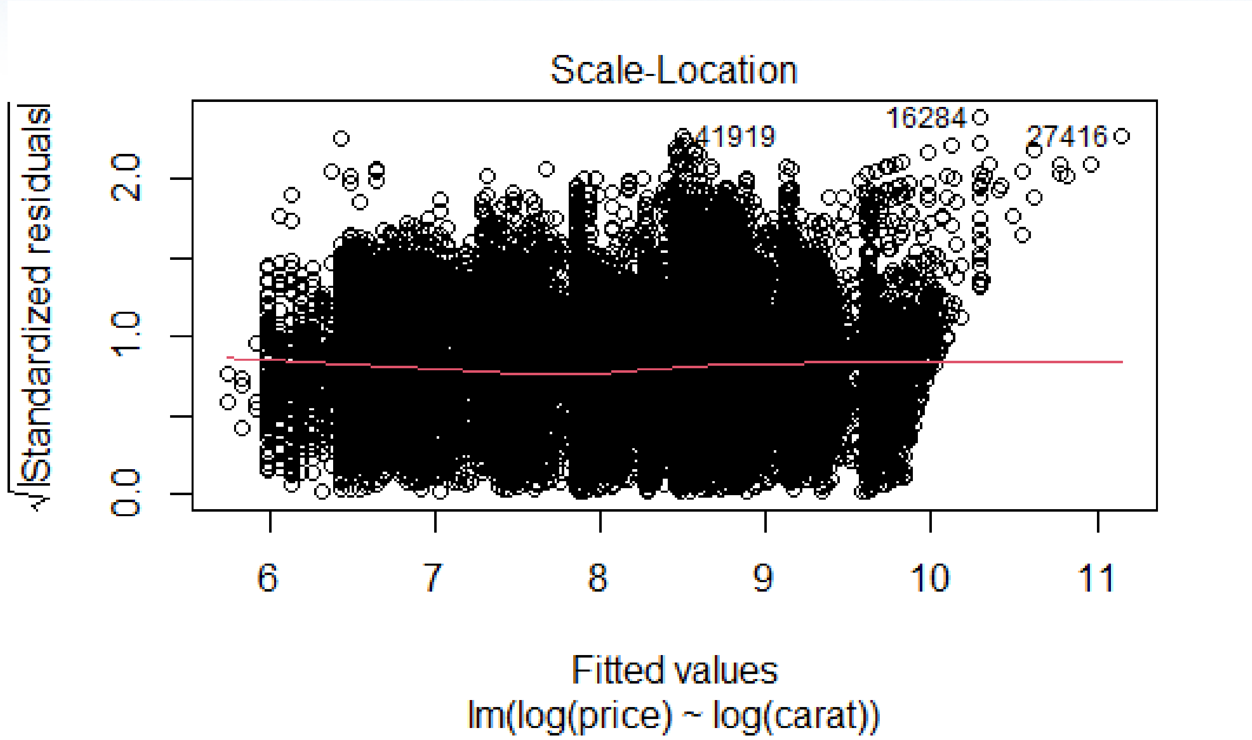
Normality → QQ plot

points should follow the diagonal dashed line.

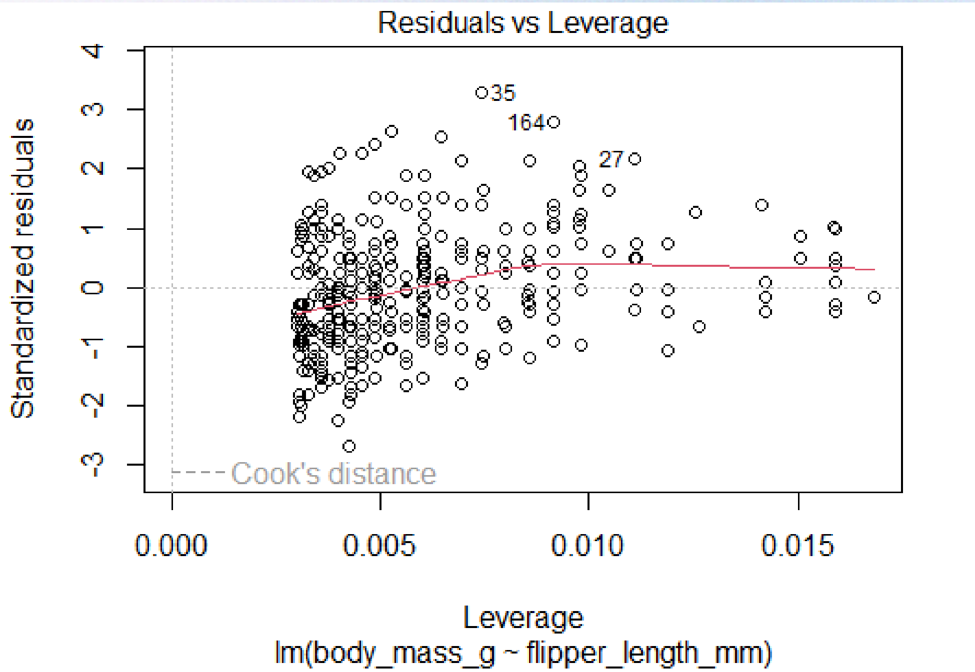


Homoscedasticity → Scale-Location

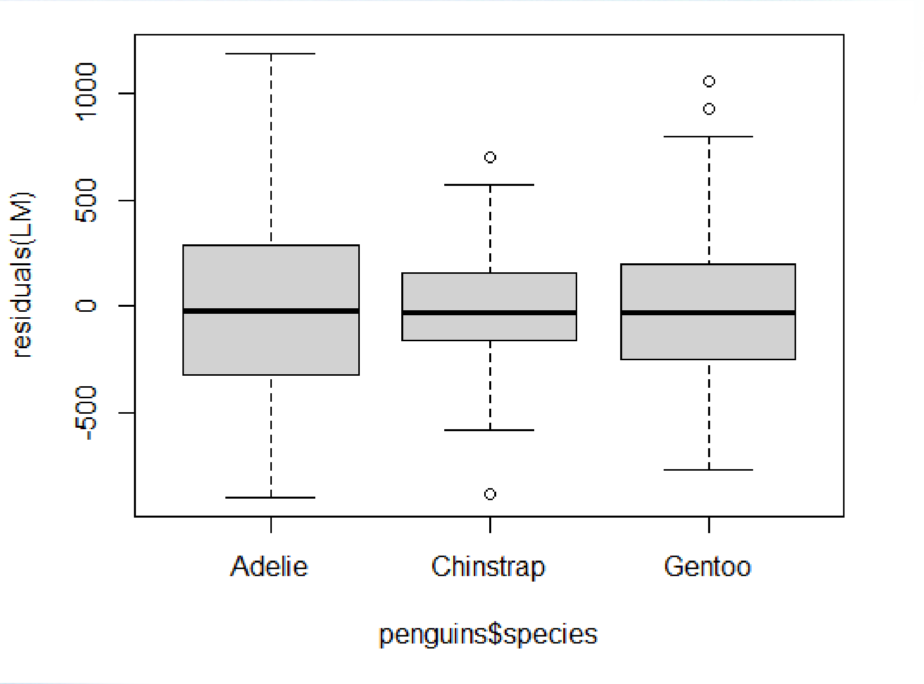
points should be spread equally around a horizontal line.

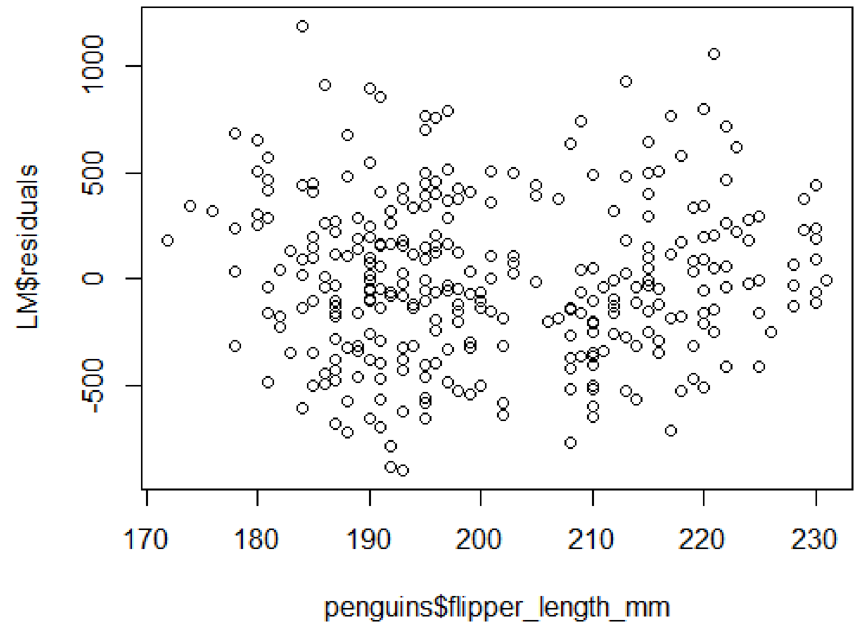


Influential observations → Residuals vs Leverage

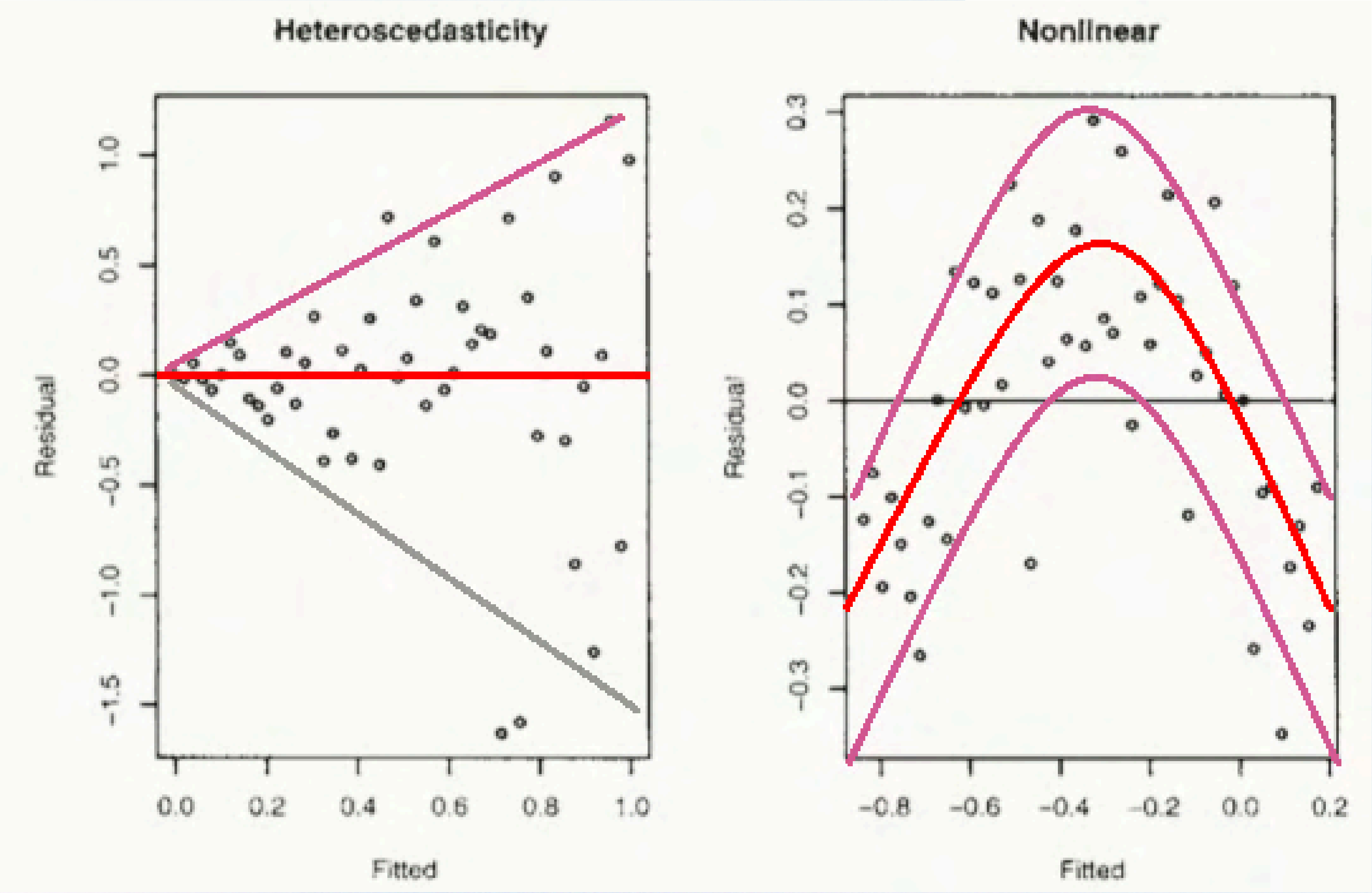


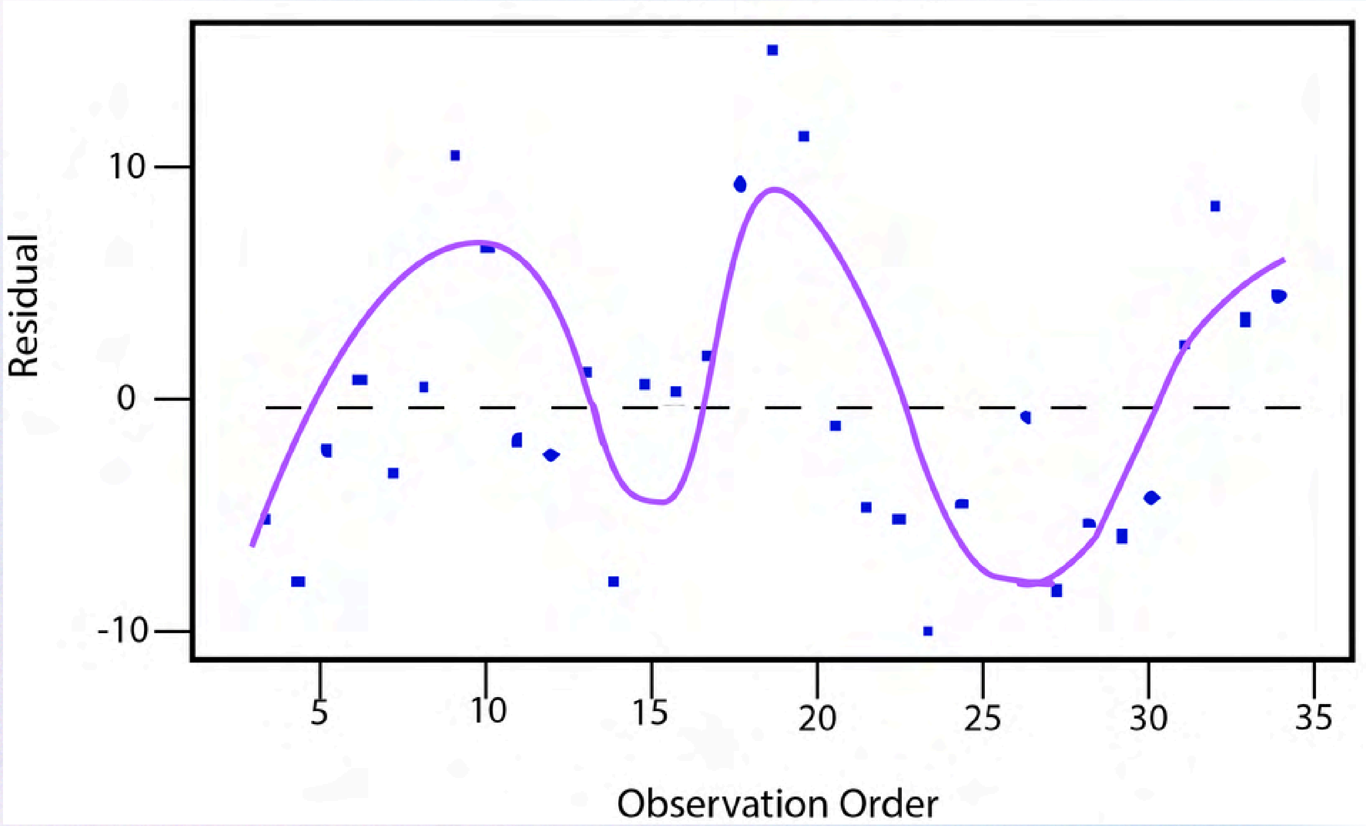
Independence → Residuals vs Each explanatory variable

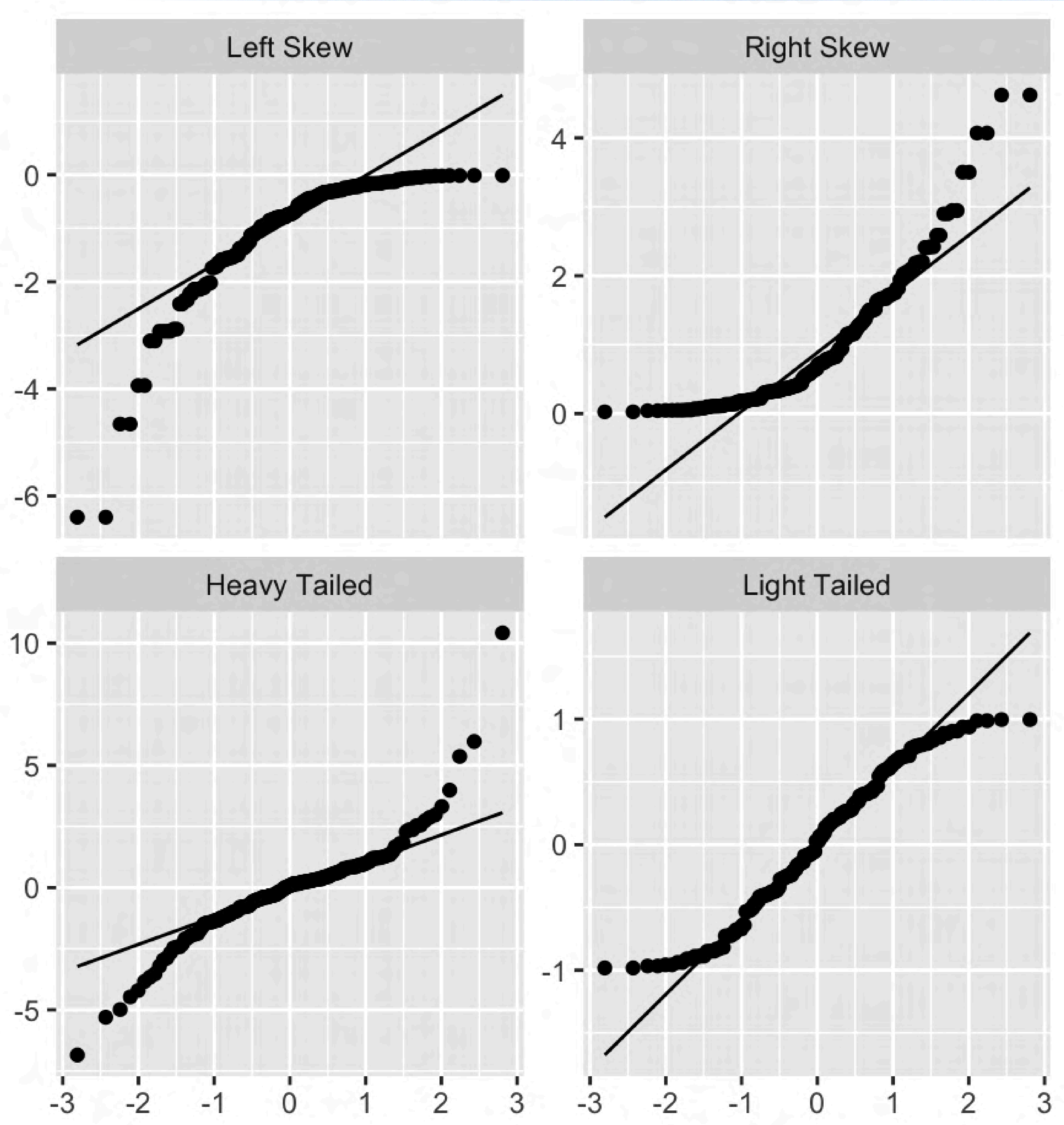




“Bad” Outputs


Heteroscedasticity
Nonlinear
Residual
Fitted
0.0
0.2
0.4
0.6
0.8
1.0
-0.8
-0.6
-0.4
-0.2
0.3
0.1
-0.1
-0.3
0.5
-0.5
-1.0
-1.5


Residual
Observation Order
10
0
-10
5
15
20
25
30
35


Left Skew
Right Skew
Heavy Tailed
Light Tailed
0
-2
-4
-6
4
2
10
5
-5
1
-1
-3
3

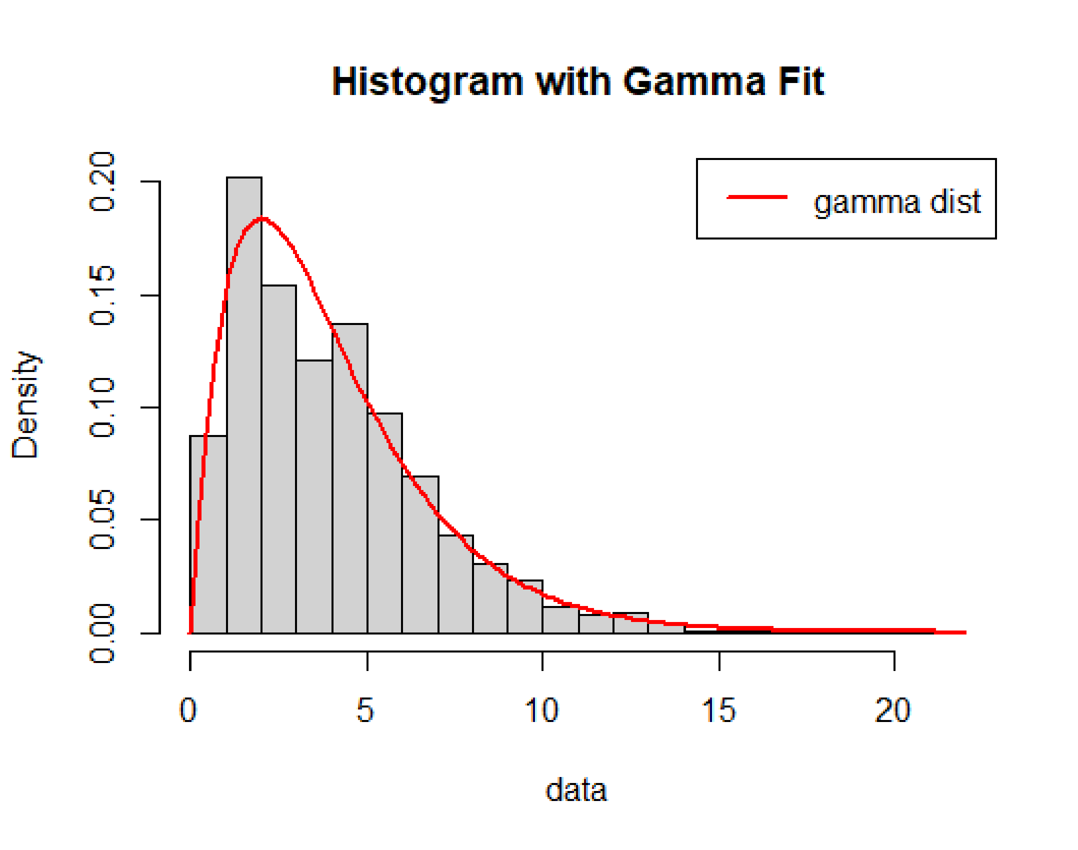
Generalized Linear Model (GLM)

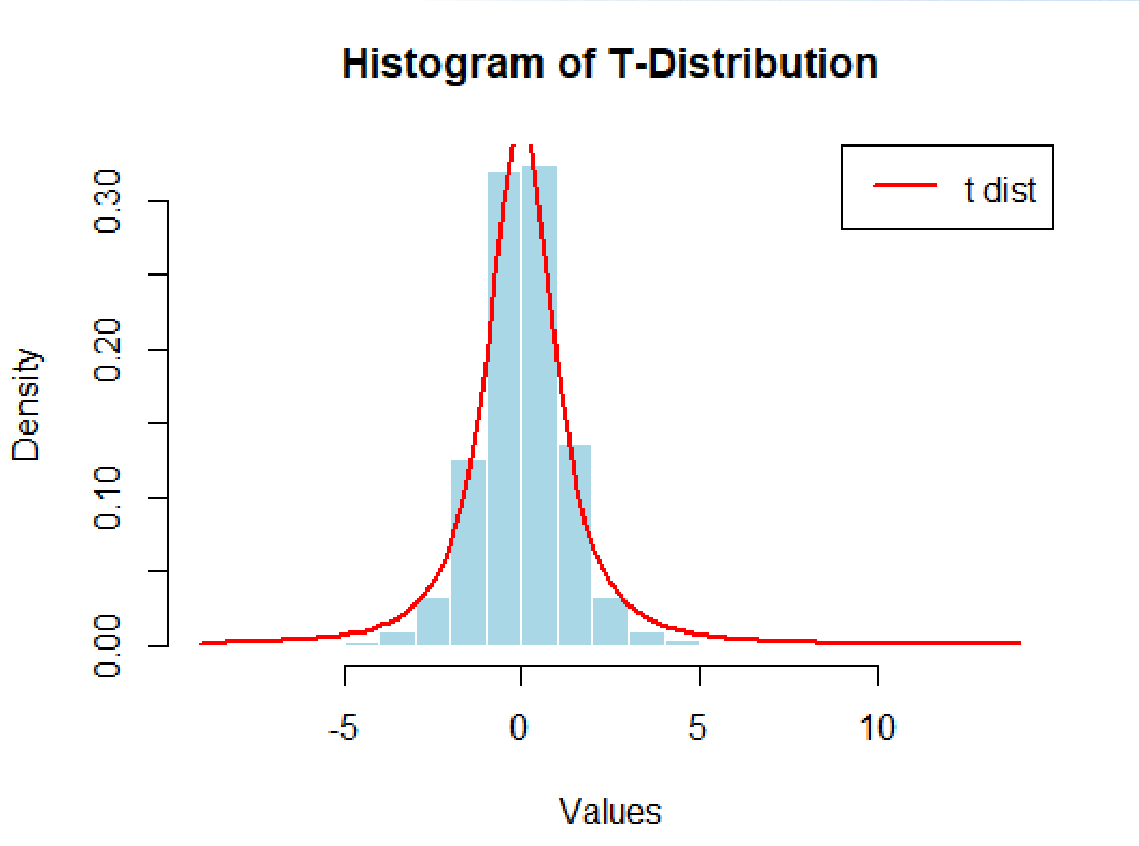
$$g(y) = X\beta + \varepsilon, y \sim E(\boldsymbol{\theta}), \beta = \mathrm{C}$$

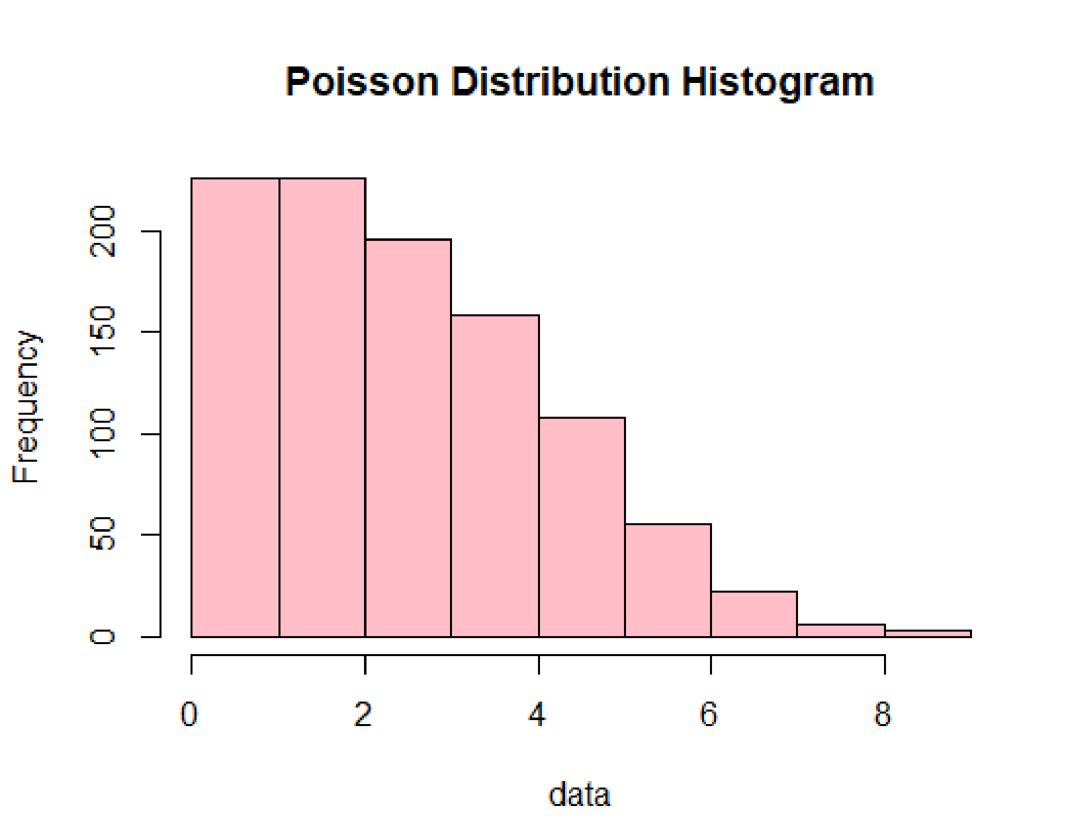
Key Assumptions

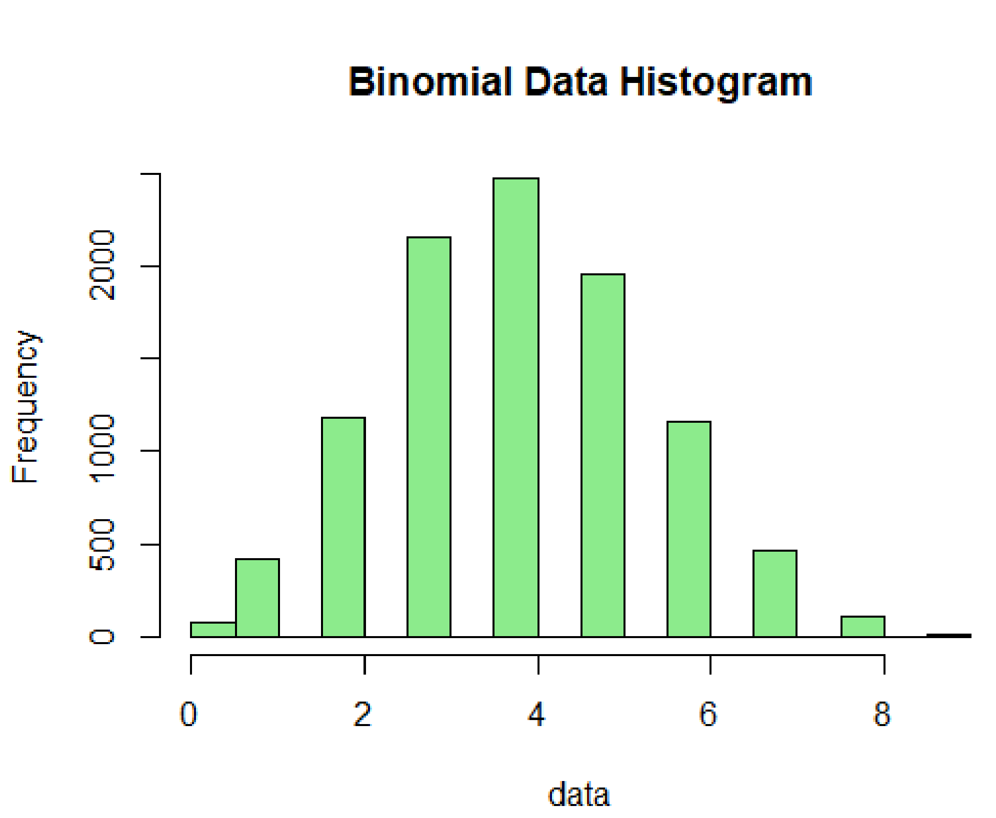
- **Correct Distribution: The response variable follows a distribution from the exponential family (e.g., Binomial, Poisson, Gamma, Normal).**
- **Linearity: The transformation of the response g(y) is linearly related to the explanatory variables βX**
- **Independence: Observations are independent.**

Exponential family distributions









Distribution selection strategy

01 Identify the Response Type

- Continuous (numeric values)
  Examples: weight, temperature

Common choices:
Normal → symmetric data
Gamma → positive & skewed
t-distribution → heavy tails
Beta → bounded (0–1)

- Discrete (counts: 0, 1, 2, ...)
  Examples: number of events

Common choices:
Poisson → mean ≈ variance
Negative Binomial → overdispersion (variance > mean)

- Categorical (classes / labels)
  1. Binary (2 categories)
     Yes/No, 0/1 → Logistic regression
  2. Multiclass (unordered)
     North, South, East, West

Common choices:
Multinomial logistic
Random forest

02 Explore the Data (EDA)

- Plot the response (e.g., histogram)
- Look at shape: symmetric, skewed, bounded, counts

Choose distributions that match the data pattern

03 Fit Candidate Models

- Try multiple reasonable distributions
  (e.g., Normal vs Gamma, Poisson vs Negative Binomial)

04 Compare & Diagnose Models

- Check residuals (patterns, variance, normality)
- Evaluate model fit

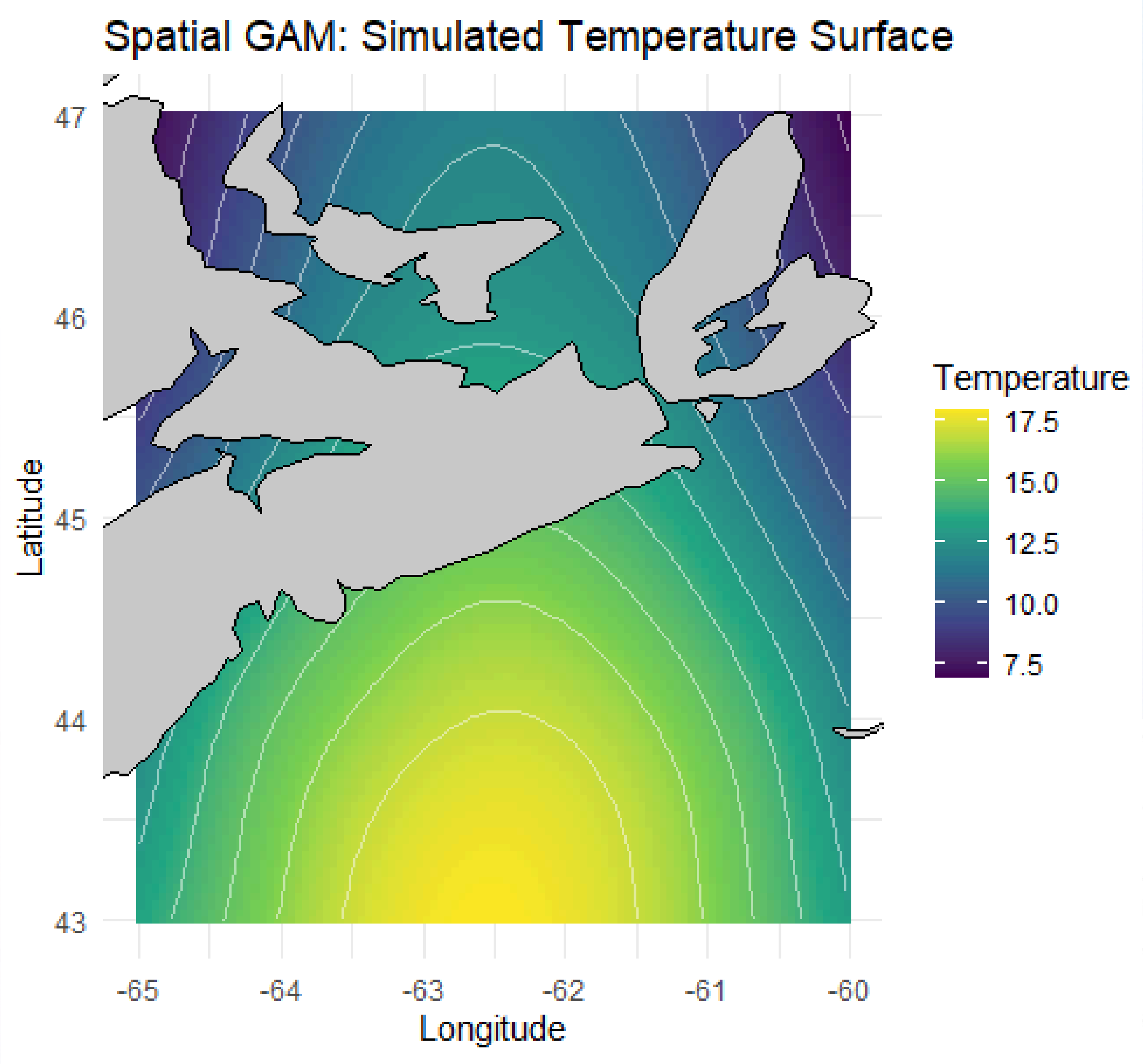
Select the model that best fits the data and assumptions

GLM Rsidual Analaysis

LM vs GLM	Randomized Quantile Residuals (RQR)	R Tools
• LM assumes normal responses (sepecial case in GLM) • GLMs allow non-normal responses (e.g., Poisson, Binomial) Standard residuals may not behave well	How it works: • Transform residuals to be approximately normal • Works across different GLM distributions Makes QQ plots meaningful again • Reference: Dunn & Smyth (1996) What to Check: • QQ plot → should follow straight line • Residual vs fitted → no clear pattern • Outliers → unusually large residuals	• statmod::qresid() → compute RQR • DHARMa→ simulation-based residual diagnostics Built-in plots Works for many GLMs / GLMMs

Generalized Additive Model (GAM)

$$g(y) = a + f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) + \varepsilon, y \sim E(\boldsymbol{\theta})$$

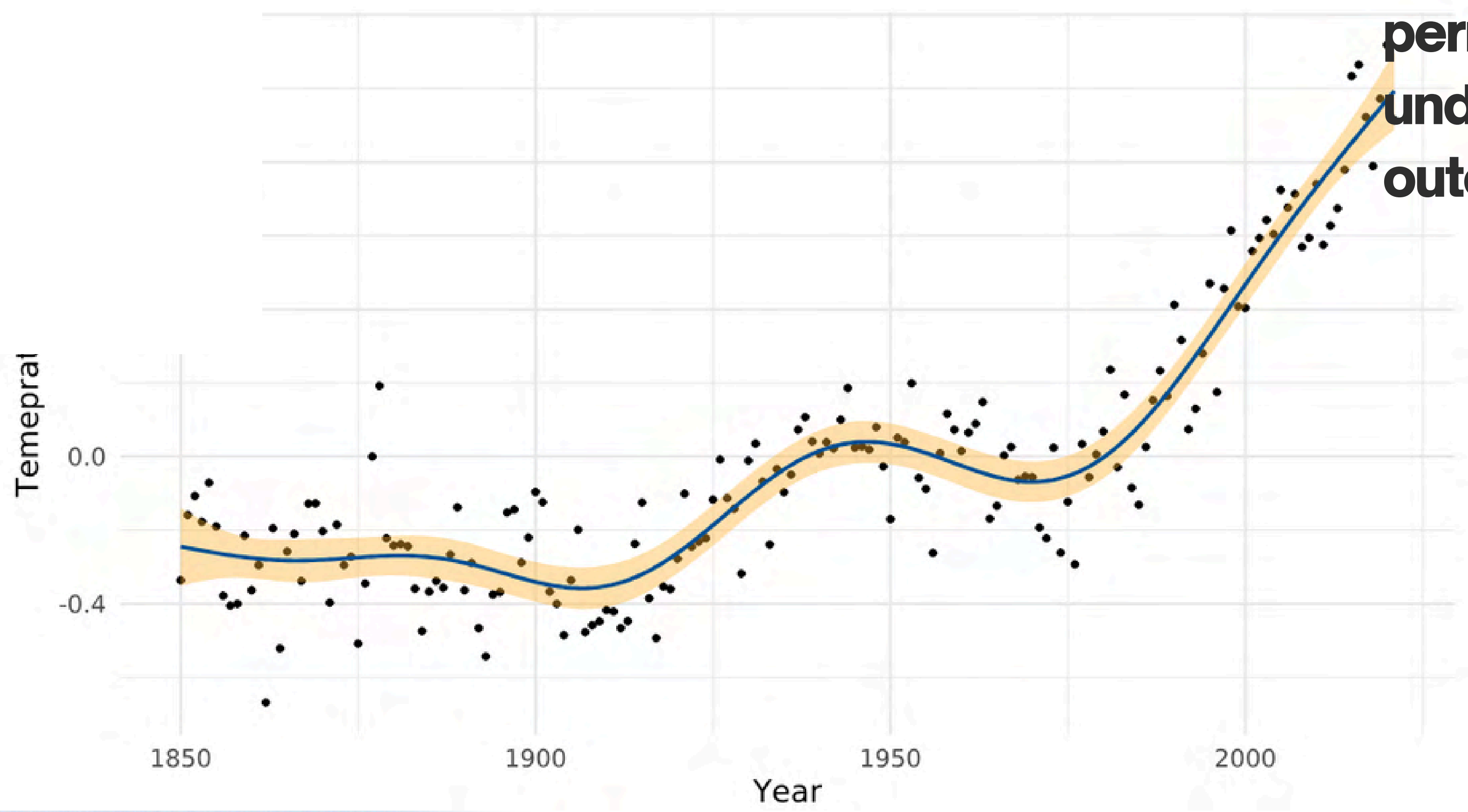


Key Assumptions

- Correct Distribution
- Independence

Key Characteristics

- Non-linear Relationships: GAMs use smoothing functions : automatically or manually detect model non-linear relationships.
- Additive Structure: The model relates the response variable to predictors through sum of smooth functions f_i(x_i)
- Flexible Distributions: Same as GLMs.
- Interpretability: Unlike "black box" models, GAMs permit inspection of each smooth function to understand how a specific predictor affects the outcome.



Smooth Functions

Function	When to Use	How to Use in R	Method (Idea)	Output Interpretation
s(x)	1D nonlinear effect	y ~ s(x)	Spline (basis expansion + penalty)	Smooth curve of x vs y
s(x, k=...)	Control flexibility	y ~ s(x, k=10)	More basis functions = more wiggle	Higher k → more flexible curve (knots and degree)
te(lon,lat)	Interaction (2D smooth)	y ~ te(lon, lat)	Tensor product smooth	Surface (lon & lat jointly affect y)
ti(x, z)	Interaction + main effects separate	y ~ s(x) + s(z) + ti(x,z)	ANOVA-style decomposition	Interaction effect only
t2(x, z)	Stable tensor smooth (advanced)	y ~ t2(x, z)	Alternative tensor basis	Similar to te() but more robust
s(x, bs="cs")	Shrinkage smooth	y ~ s(x, bs="cs")	Penalized spline with shrinkage	Can shrink to zero effect
s(x, bs="re")	Random effect	y ~ s(group, bs="re")	Random intercepts	Group-level variation

LM vs GLM vs GAM — Summary

Model	When to Use	Key Assumptions	R Package / Function	What to Check
LM	Continuous response, linear relationship	Linearity Normality Homogeneity Independence	stats::lm()	Residual vs fitted QQ plot Scale-Location Residuals vs each explanatory variable Residuals vs Leverage
GLM	Non-normal response (binary, counts)	Linearity Independence	stats::glm()	Use RQR to check everything above Check overdispersion for Poisson distributions
GAM	Nonlinear relationships	Independence	mgcv::gam()	Smooth plots RQR diagnostics Check k / wiggliness for overfitting

When Independence Assumption Breaks

Hospital Name	Patient ID	Treatment	Dosage	Hospitalization duration	Disease Type	Sex
QII	1	A	75 ug	1 month	Disease X	F
QII	2	A	70 ug	1 month	Disease X	F
QII	3	B	80 ug	1.5 month	Disease X	F
QII	4	B	40ug	2 month	Disease X	M
QII	5	C	50 ug	0.5 month	Disease X	M
IWK	6	A	75 ug	1 month	Disease X	F
IWK	7	A	70 ug	1 month	Disease X	F
IWK	8	C	80 ug	0.5 month	Disease X	F
IWK	9	B	40ug	2 month	Disease X	M
IWK	10	A	50 ug	3 month	Disease X	M

LM / GLM / GAM assume:

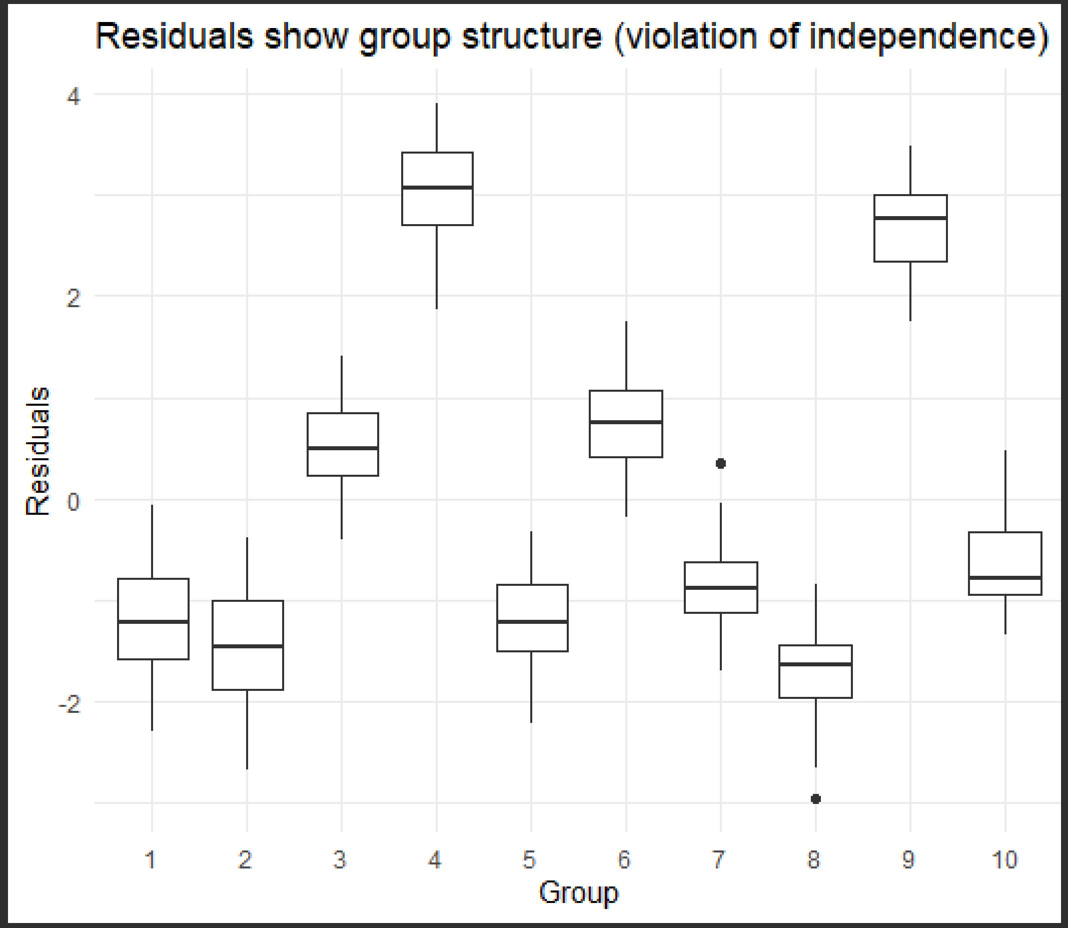
Observations are independent

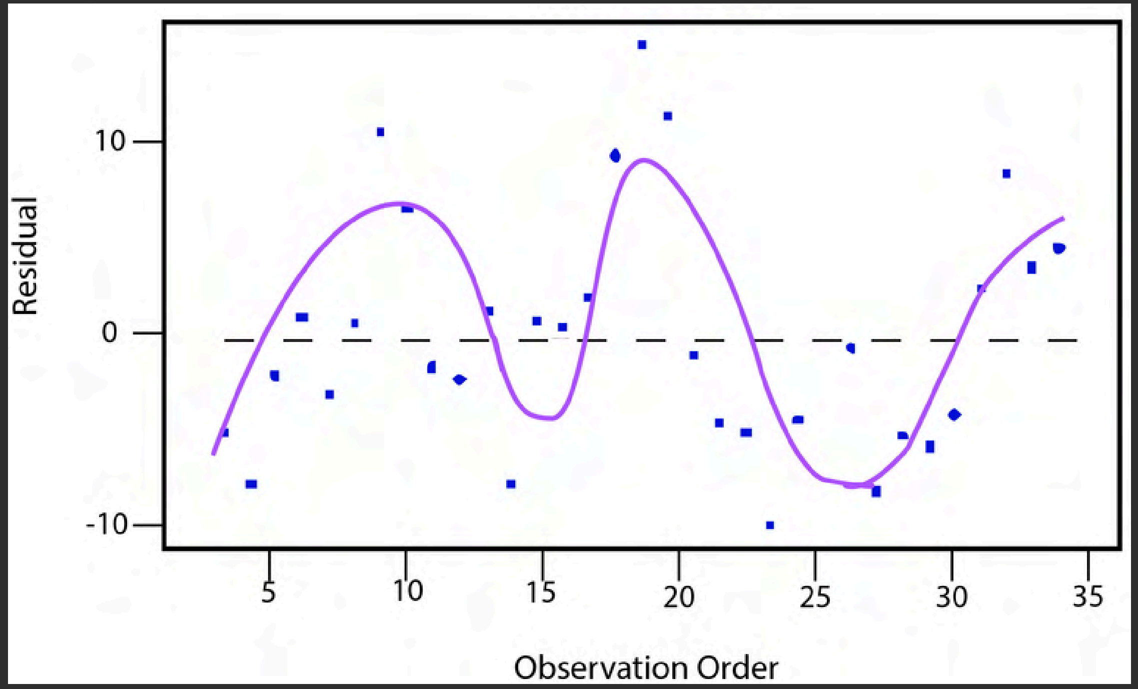
Residuals show no pattern

But some data are grouped or correlated:

- Medical trials → patients within hospitals
- Spatial data → nearby locations are similar
- Time series data → autocorrelation
- Engineering data→ repeated measures / sites

What happens if we ignore this?





- Residuals show patterns
- Standard errors are wrong
- Inference becomes unreliable

Modeling Group Structure

Fixed Effects

Estimate specific differences between observed groups

Use when:

- Interested in each level explicitly
- Small, meaningful categories

Random Effects

Model variation across groups

Use when:

- Data has grouping / clustering structure
- Want to generalize beyond observed levels
- Many groups / repeated measures

Choose Random Effects over Fixed Effects

- Residual patterns by group → clustering, non-random structure
- Repeated measurements within groups→ same subject, site, year
- Hierarchical data structure→ observations nested in groups
- Correlation within levels→ observations not independent

Same Variable, Different Modeling Choices

Year as fixed effect

compare specific years → dummy variable

Year as random effect

treat years as sampled → estimate variability/correlation across years → time series models

Space *(lon, lat)*

GAM: smooth function *f(lon,lat)*

GLMM: spatial correlation

Choice depends on the question, not the variable itself

Advantages of Random Effects:

Handle correlation within groups

Accounts for non-independence (core reason)

More efficient with many groups

Avoid estimating a parameter for each level

Enable generalization

Inference extends beyond observed groups

Partial pooling (shrinkage)

Stabilizes estimates, especially with small sample sizes

Generalized Linear Mixed Model (GLMM)

$$g(y) = X\alpha + Z\beta + \varepsilon, y \sim E(\boldsymbol{\theta}), \beta \sim \Omega$$

What each part means

y → response variable
Response follows an exponential family distribution (Normal, Binomial, Poisson, etc.)

g(.)→ link function
(connects mean of y to predictors)

Xα → fixed effects
(population-level effects, like LM/GLM)

Zβ→ random effects
(group-level variation, e.g., site, patient)
Random effects follow a distribution (typically Normal with mean 0)

ϵ→ residual error

Key Assumptions

- Observations are independent conditional on random effects
- Random effects are normally distributed
- Correct link function & response distribution
- Model captures group structure / correlation

What to say:

- GLMM extends GLM by adding random effects
- It allows us to model grouped or correlated data
- Independence is not removed—it's handled through random effects

GLMM = GLM + Random Effects

Generalized Additive Mixed Model (GAMM)

$$g(y) = a + f(x) + Z\beta + \varepsilon, y \sim E(\boldsymbol{\theta}), \beta \sim \Omega$$

What GAMM adds

- Captures nonlinear relationships
- Keeps group structure (random effects)
- Combines flexibility + hierarchy

Practical Considerations

More flexible → higher risk of overfitting

Requires careful:

- smoothness selection
- diagnostics

Interpretation becomes less straightforward

With flexibility comes responsibility

GAMM opens the door to advanced models:

- Spatiotemporal models
  spatial + temporal smooths
- State-space models
  latent processes over time
- Time-series models
  autocorrelation + trends

Spatiotemporal , State-space, Time-series models can often be formulated within the GLMM/GAMM framework.

If you are interested in learning these models, let us know, I'll be willing to teach

Random Effects Structure in GLMM

Structure	What it assumes	When to use	R formula (glmmTMB)
Random Intercept + Slope	Groups differ in baseline *and* trend, and they are related	Most flexible / default assumption	glmmTMB(y ~ x + (x | group), data = df)
Random Intercept only	Groups differ in baseline but are the same in trend	Groups start differently, but respond to x in the same way	glmmTMB(y ~ x + (1 | group), data = df)
Random Slope only	Same baseline across groups, but different trends	When groups start similarly but evolve differently	glmmTMB(y ~ x + (0 + x | group), data = df)

Random Effects Examples

1. Random Intercept Only

Example: Hospital patient recovery
y: recovery time
X: treatment dose
Group: hospital

Interpretation:
Hospitals differ in baseline recovery (some are generally faster)
BUT treatment effect is the same everywhere

```r
library(glmmTMB)

model_ri <- glmmTMB(
  recovery ~ dose + (1 | hospital),
  data = df
)
```

2. Random Slope Only

Example: Fertilizer effect in identical greenhouses
y: plant growth
X: fertilizer amount
Group: greenhouse

Interpretation:
All greenhouses start at same baseline
BUT respond differently to fertilizer

```r
model_rs <- glmmTMB(
  growth ~ fertilizer + (0 + fertilizer | greenhouse),
  data = df
)
```

3. Random Intercept + Slope

Example: Salamander abundance in ponds
y: salamander count
X: temperature
Group: pond / site

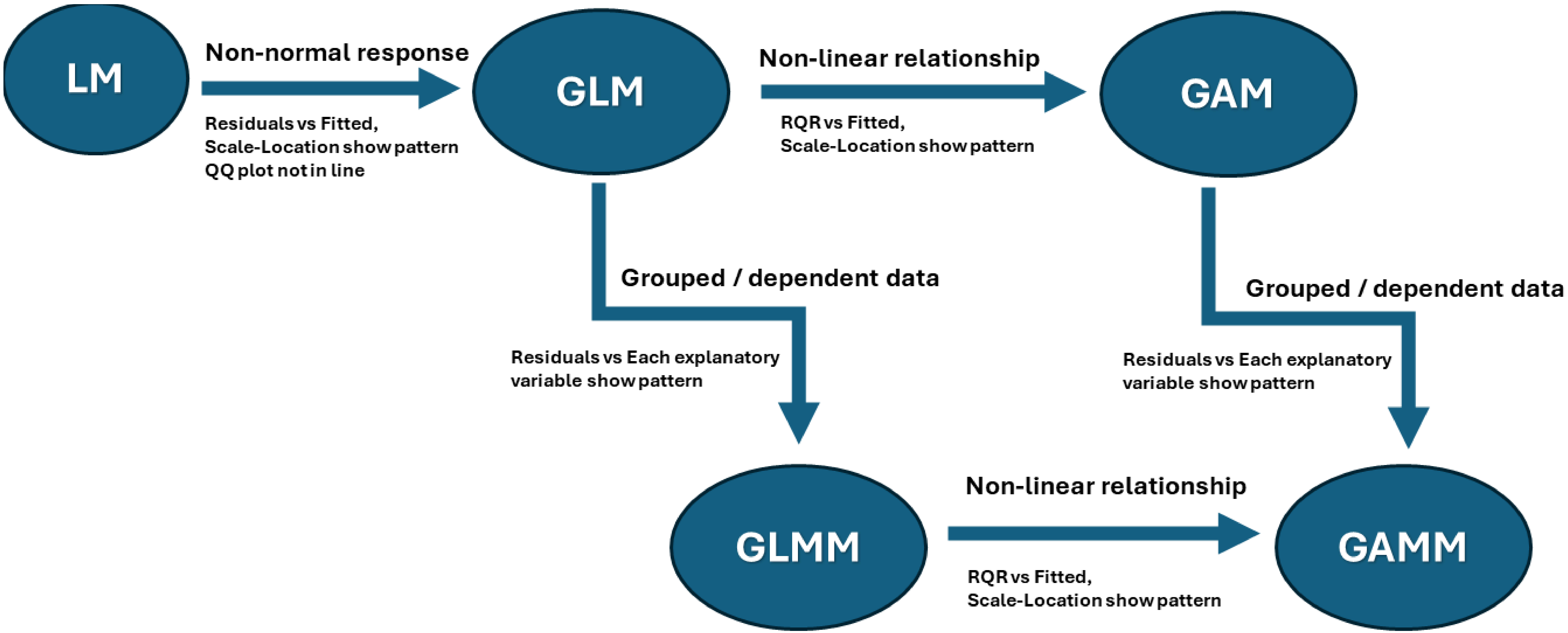
Interpretation:
Some ponds have more salamanders overall
AND temperature effect differs by pond

```r
model_ris <- glmmTMB(
  count ~ temperature + (temperature | pond),
  data = df
)
```

R Packages for GLMM / GAMM

Package	Main Use Case	Model Function	Prediction	Residuals	Optimization / Engine	What to Check After Fit
lme4	Classic GLMMs (no spatial structure)	lmer(), glmer()	predict()	residuals()	Laplace / (P)QL	Residual vs fitted, QQ plot
glmmTMB	Flexible GLMMs (Poisson, NB, zero-inflation)	glmmTMB()	predict()	residuals()	TMB (fast C++ optimization)	RQR / DHARMa diagnostics
mgcv	GAM / GAMM (smooth effects)	gam(), gamm()	predict.gam()	residuals()	Penalized likelihood (REML/GCV)	Smooth plots, k-index, residuals
sdmTMB	Large-scale spatial-temporal	sdmTMB()	predict()	residuals()	SPDE / GMRF (TMB backend)	Spatial sd error maps
staRVe	Large-scale spatial-temporal	strv_fit()	strv_predict()	residuals()	NNGP (TMB backend)	Spatial sd error maps
momentuHMM	Animal movement / state-switching models	fitHMM()	predict()	pseudo-residuals	EM algorithm	State decoding, residual autocorrelation

Evolution of Statistical Models



Thank You

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https://github.com/joyliujoyliu

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